# BIHAR ENGINEERING UNIVERSITY, PATNA

## DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

**B.Tech 4th Semester • Examination Resource**

# ANALOG ELECTRONICS & INTEGRATED CIRCUITS (2024)

**Exam Date: Saturday, 31 Oct 2026 (2:00 PM - 5:00 PM)**

### Included Question Papers

| Year | Paper Title | Course Code | Examination Session |
|---|---|---|---|
| **2024** | Analog Circuits | 104401 | B.Tech 4th Semester Examination, 2024 |

**Apna BEU Question Bank Archive** • Compiled for B.Tech ECE 4th Semester

Contains official previous year questions with multiple-choice questions, descriptive problems, and complete circuits.

# BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Examination, 2024

## ANALOG CIRCUITS

Time: 03 Hours | Code: 104401 | Full Marks: 70

**Instructions:**

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

**Q.1 Choose the correct option/answer the following (Any seven question only):** **[14]**

1. JFET can be operated with
   - i. negative gate only
   - ii. gate voltage only
   - iii. positive as well as negative gate voltage
   - iv. none of the above

2. Which power amplifier has the highest collector efficiency
   - i. C
   - ii. A
   - iii. B
   - iv. AB

3. Which amplifier is used for 180-degree phase shift
   - i. differential amplifier
   - ii. non inverting amplifier
   - iii. inverting amplifier
   - iv. None of the above

4. A MOSFET has three terminals, namely
   - i. cathode, anode, grid
   - ii. source, gate, drain
   - iii. emitter, base, collector
   - iv. none of the above

5. The maximum efficiency of resistance loaded class B power amplifier is
   - i. 5%
   - ii. 50%
   - iii. 30%
   - iv. None of the above

6. A JFET is also called _____ transistor
   - i. unipolar
   - ii. bipolar
   - iii. injunction
   - iv. none of the above

7. In a phase shift oscillator, the frequency determining elements are
   - i. L and C
   - ii. R, L and C
   - iii. R and C
   - iv. None of the above

8. The leakage current across a pn junction is due to
   - i. Minority carriers
   - ii. Majority carriers
   - iii. Junction capacitance
   - iv. None of the above

9. An oscillator converts
   - i. a.c. power into d.c. power
   - ii. a.c. power into a.c. power
   - iii. mechanical power into a.c. power
   - iv. none of the above

10. In a transistor
    - i. $I_E = I_C - I_B$
    - ii. $I_B = I_C + I_E$
    - iii. $I_C = I_E + I_B$
    - iv. $I_E = I_C + I_B$

**Q.2 Solve both questions :** **[14]**

1. In a full wave rectifier, the input is from 30-0-30V transformer. The load and diode forward resistances are 100Ω and 10 Ω respectively. Calculate the average voltage, dc output power, ac input power, rectification efficiency and percentage regulation.
2. Explain the working of positive clamping circuit.

**Q.3 Solve both questions :** **[14]**

1. Draw the common emitter circuit and sketch the input and output characteristics. Also explain active region, cutoff region and saturation region by indicating them on the characteristic curve.
2. With a neat circuit diagram explain the Voltage Divider Bias circuit by giving its exact analysis.

**Q.4 Solve both questions :** **[14]**

1. Explain the characteristics of an Ideal Op-Amp. Mention some of the applications of Op-Amp.
2. Explain R-2R ladder DAC with example?

**Q.5 Solve both questions :** **[14]**

1. Explain Successive type ADC with neat circuit diagram.
2. Describe the construction and operation of JFET and draw its characteristics.

**Q.6 Solve both questions :** **[14]**

1. Explain Colpitts oscillator with Circuit diagram and derive its frequency.
2. Explain RC coupled transistor amplifier with neat circuit diagram in a middle frequency Range.

**Q.7 Solve both questions :** **[14]**

1. Explain RC- phase shift oscillator using op-amp and derive its frequency.
2. Describe the construction and operation of depletion type MOSFET and draw its characteristics.

**Q.8 Solve both questions :** **[14]**

1. Describe class A Power Amplifier. Derive the expression for its maximum conversion Efficiency.
2. Explain Wien bridge oscillator using op-amp and derive its frequency.

**Q.9 Solve both questions :** **[14]**

1. Explain MOSFET as an amplifier?

2. Describe class B Power Amplifier. Derive the expression for its maximum conversion Efficiency.